VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Experiment No. 08

Aim: - Write a Java program to implement user defined packages in terms of creating a new package.

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

Theory: -

Packages

Packages in Java are a mechanism to encapsulate a group of classes, interfaces, and sub-packages. In Java, it is used for making search/locating and usage of classes, interfaces, enumerations, and annotations easier. It can be considered data encapsulation also. In other words, we can say a package is a container of a group of related classes where some of the classes are accessible are exposed, and others are kept for internal purposes.

Types of Packages

Packages are categorized into two parts. These are:

- **Built-in packages:** The already defined package like java.io.*, java. lang.* etc., are known as built-in packages.
- **User-defined packages:** As the name propose, user-defined packages in Java are essentially packages that are defined by the programmer. Whenever we want to add a class to the package, we have to mention the package name and the "package" keyword at the top of the program.

User-defined Packages

User-defined packages are those packages that are designed or created by the developer to categorize classes and packages. They are much similar to the built-in that java offers. It can be imported into other classes and used the same as we use built-in packages. But If we omit the package statement, the class names are put into the default package, which has no name.

To create a package, we're supposed to use the package keyword.

Syntax:

package package-name;

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Steps to create User-defined Packages

Step 1: Creating a package in java class. The format is very simple and easy. Just write a package by following its name.

```
package example1;
```

Step 2: Include class in java package, but remember that class only has one package declaration.

```
package example1;

class gfg {
  public static void main(Strings[] args){
    -------function--------
  }
}
```

Step 3: Now the user-defined package is successfully created, we can import it into other packages and use its functions it.

Advantages of Packages:

- **Re-usability:** The classes contained in the packages of another program can be easily reused.
- **Name Conflicts:** Packages help us to uniquely identify a class.
- **Controlled Access:** Offers access protection such as protected classes, default classes and private class.
- **Data Encapsulation:** They provide a way to hide classes, preventing other programs from accessing classes that are meant for internal use only.
- **Maintenance:** With packages, you can organize your project better and easily locate related classes.

Disadvantages of Packages

- **Increased Complexity:** Managing multiple packages in large projects can be difficult. Requires proper structuring and maintenance.
- **Name Conflicts:** Even though packages help avoid name conflicts, different packages may still have classes with the same name, leading to confusion. Explicit package imports are needed to differentiate them.

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- **Access Restrictions:** Classes and members with default access (package-private) cannot be accessed outside the package. If a class is declared private or has limited access, it cannot be reused easily in other packages.
- **File Management Overhead:** Java requires a strict directory structure that matches the package name. Moving or renaming packages means adjusting multiple file paths.

Program and Output:

Step 1: Create the package
Save this file as MyPackageClass.java inside a folder named mypackage.

```
// File: MyPackageClass.java (inside 'mypackage' folder)
package mypackage;

public class MyPackageClass {
    public void displayMessage() {
        System.out.println("Hello from MyPackageClass in mypackage!");
    }
}
```

Step 2: Use the package in another class
Save this file as MainApp.java in the same directory where mypackage folder exists.

```
// File: MainApp.java
import mypackage.MyPackageClass;

public class MainApp {
    public static void main(String[] args) {
        MyPackageClass obj = new MyPackageClass();
        obj.displayMessage();
    }
}
```

Output:

```
Hello from MyPackageClass in mypackage!
```

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Conclusion: Hence, in this practical, we learned and implement user defined packages in terms of creating a new package the in Java Successfully.

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